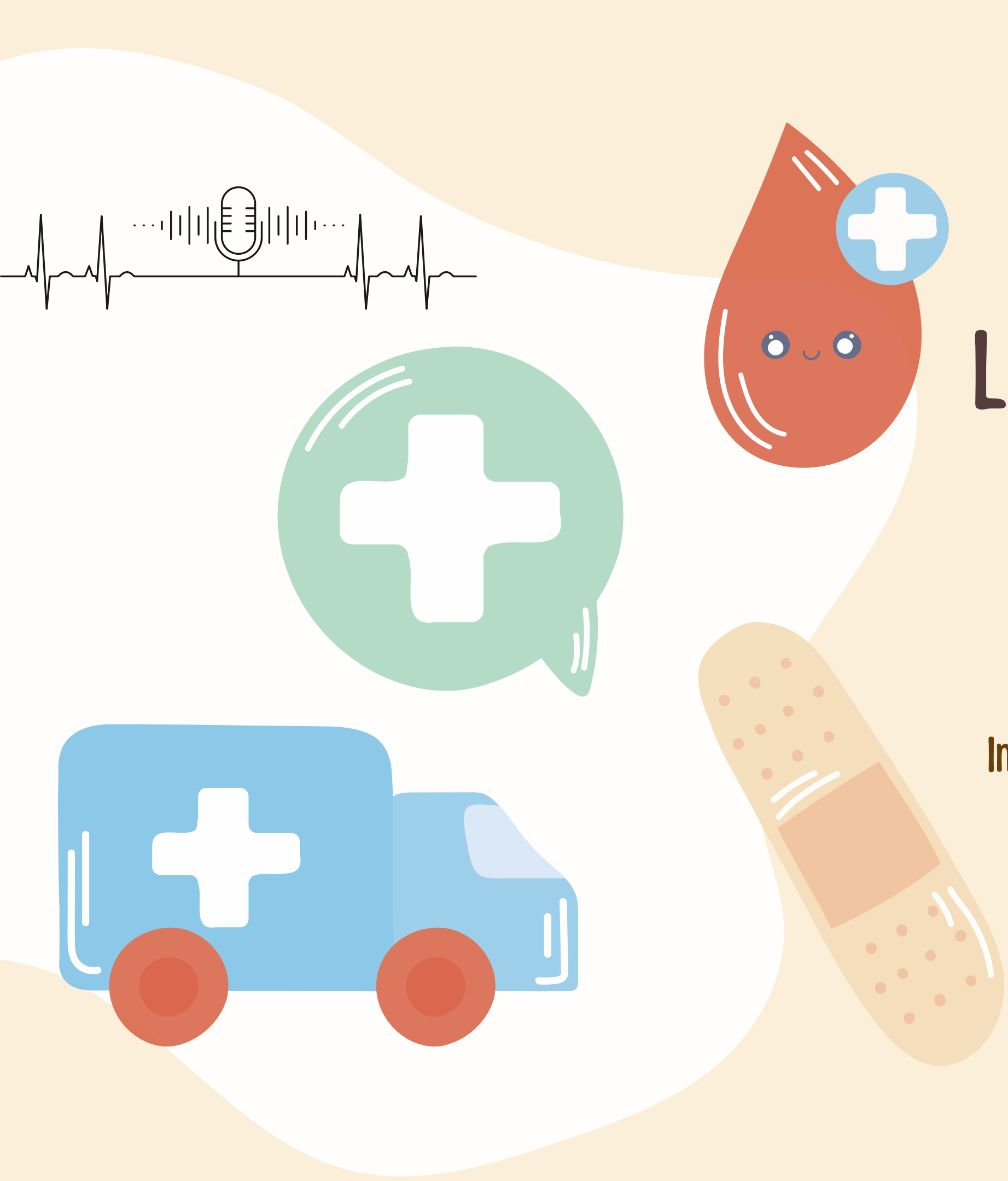




MedAlert

LLM-BASED EMS VOICE REPORT CLASSIFICATION FOR ED DECISIONS

Improving Robustness to ASR Noise with Clean and Noisy Training

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PROJECT DEFINITION

MOTIVATING USE CASE

EMS pre-arrival reports provide early clinical information that may help the ED prepare the appropriate care area and specialty consultation.

However, these reports are often brief, incomplete, and communicated verbally under noisy ambulance conditions.

To simulate this setting, we converted the reports to speech, added ambulance noise, and transcribed them using ASR.

PROBLEM DEFINITION

Task:

Predict ED preparation decisions from EMS pre-arrival handoff reports.

input:

Synthetic EMS pre-arrival handoff report

Outputs:

1. ED care area (arrival_destination)
2. Primary specialty consultation (specialty_consult_label)

Main research question:

How well can ED decisions be predicted from EMS reports, and does noisy ASR training improve performance?

Models:

DistilBERT and BioClinicalBERT

NOVELTY

We generate synthetic noisy EMS training data and train one Transformer model directly on the full report.

**WE DO NOT USE A TRADITIONAL ASR → INFORMATION
EXTRACTION →
RULE-BASED DECISION PIPELINE.**

**The model learns from both the clinical context and
recurring ASR errors.**

**We test whether noisy and clean+noisy training
improve downstream ED decision prediction.**

DATASET PREPROCESSING

- **Source:** MIMIC-IV-Ext CDS task-specific files merged by stay_id.
- **Files used:** triage_level.csv, diagnosis.csv, and specialty_referral.csv.
- **Working subset:** 2,139 complete cases with HPI, initial vitals, triage, diagnosis, and specialty referral.

Cleaned noisy diagnosis/specialty
fields:

specialties 591 → 63,
diagnoses 1,886 → 1,797

Parsed initial vitals from text into numeric
variables and removed clinically implausible
values.

Labeling

FIRST LABEL: ED CARE AREA

- Created using rule-based weak labeling.
- Input signals for labeling: clinical narrative, triage level, and parsed vital signs.
- Rules identify clinical instability, isolation risk, trauma/high-risk injury patterns, and low-acuity presentations.
- Used as target only; excluded from the text-generation prompt.

6 Final classes:

standard_ed_bed, monitored_bed, resuscitation_bay,
isolation_room, trauma_bay, triage_or_waiting_room

TARGET LABEL DISTRIBUTION AND CLASS IMBALANCE



Labeling

SECOND LABEL: PRIMARY SPECIALTY CONSULTATION

- Derived from the cleaned specialty field.
- Original specialties were grouped into broader clinical consult labels.
- 63 original specialties were reduced to **18 final consult classes**.
- Generic labels such as medicine_or_ed_consult and other_consult were used only when no specific consult label was available.
- Used as target only; excluded from the text-generation prompt.

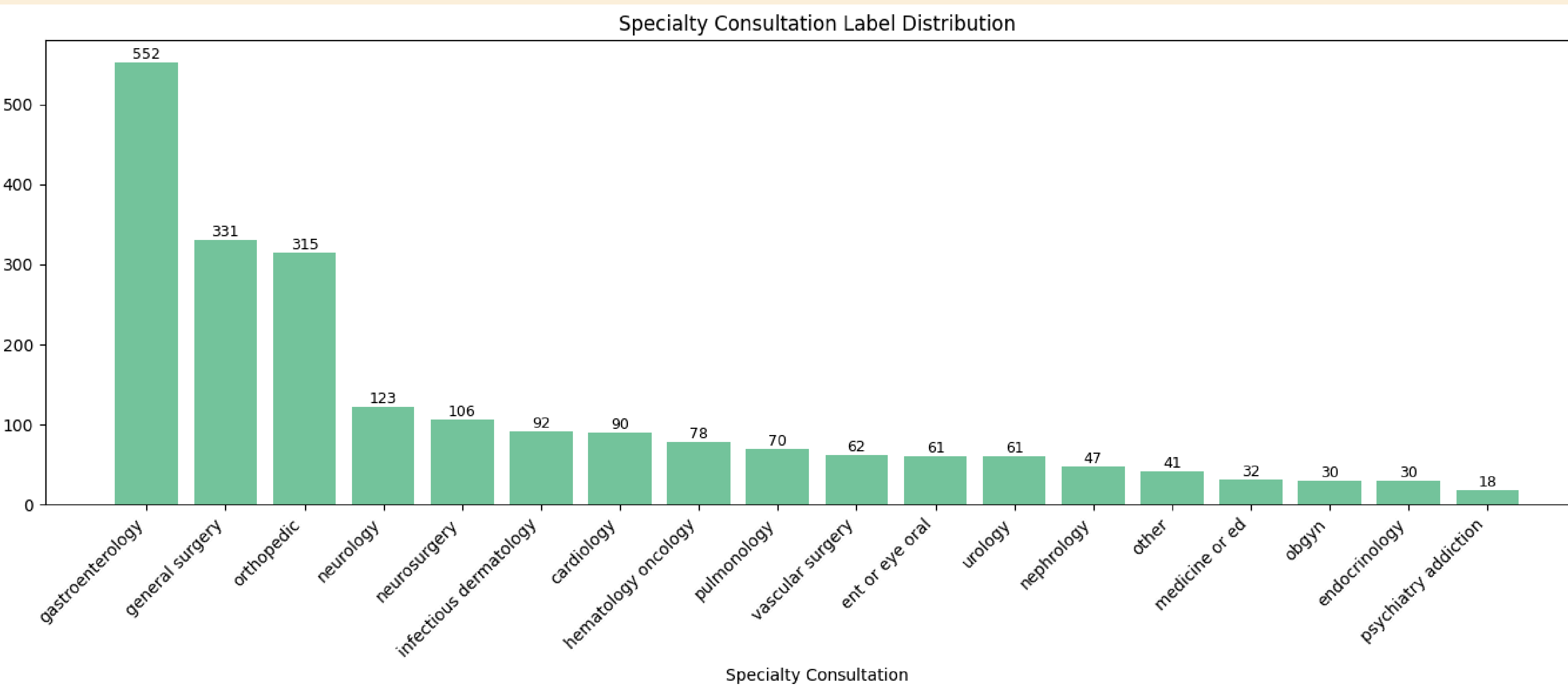
Examples:

orthopedics / hand surgery / podiatry → orthopedic_consult

ENT / ophthalmology / oral surgery / dentistry → ent_or_eye_oral_consult

**We intentionally prioritize ASR robustness over multi-label modeling in order to keep the experimental question focused and interpretable.*

TARGET LABEL DISTRIBUTION AND CLASS IMBALANCE



LEAKAGE-AWARE CASE SEED FOR REPORT GENERATION

- Each case was converted into a structured seed containing HPI, patient information, and initial vital signs.
- To reduce leakage, the seed excluded target labels, specialty fields, diagnosis fields, and triage, since these were related to the target-label creation.

```
{
  "source_case_id": 32822973,
  "patient_info": "Gender: Female, Race: WHITE, Age: 80",
  "hpi": "Patient is an ___ year-old patient with history of Sjogren's\nsyndrome, moderate MR, recent hospi
  "initial_vitals_raw": "Temperature: 100.2, Heartrate: 95.0, resprate: 28.0, o2sat: 100.0, sbp: 99.0, dbp:
  "vitals": {
    "temperature": 100.2,
    "heart_rate": 95.0,
    "respiration_rate": 28.0,
    "oxygen_saturation": 100.0,
    "systolic_bp": 99.0,
    "diastolic_bp": 47.0
  }
}
```

SYNTHETIC EMS REPORT GENERATION

2,139 LABELED CLINICAL CASES



LEAKAGE-AWARE CASE SEEDS



GPT-4.1-MINI TEXT GENERATION



4 EMS REPORT VARIANTS PER CASE



8,556 SYNTHETIC EMS REPORTS

4 VARIANTS PER CASE:

1. professional_complete
2. brief_radio_missing_details
3. patient_reported_uncertain
4. distracted_or_disorganized_handoff



FINAL DATASET LABEL DISTRIBUTION

**The generated dataset preserves the original class imbalance, since each source case produced four report variants.*

Specialty consult label distribution:

specialty_consult_label	
gastroenterology_consult	2208
general_surgery_consult	1324
orthopedic_consult	1260
neurology_consult	492
neurosurgery_consult	424
infectious_dermatology_consult	368
cardiology_consult	360
hematology_oncology_consult	312
pulmonology_consult	280
vascular_surgery_consult	248
ent_or_eye_oral_consult	244
urology_consult	244
nephrology_consult	188
other_consult	164
medicine_or_ed_consult	128
obgyn_consult	120
endocrinology_consult	120
psychiatry_addiction_consult	72

Arrival destination distribution:

arrival_destination	
standard_ed_bed	4216
monitored_bed	3476
resuscitation_bay	580
isolation_room	192
triage_or_waiting_room	68
trauma_bay	24

Example: Same Clinical Case, Different EMS Report Variants

Source case ID: 30001856

Arrival destination: standard_ed_bed

Specialty consult label: gastroenterology_consult

Text variant: professional_complete

We are en route with a 54-year-old male with a history of alcoholic cirrhosis and hepatitis C, status post TIPS procedure, presenting with three weeks of worsening epigastric pain now constant and radiating around his abdomen. He reports vomiting this morning, possibly with blood, and subjective fever and chills earlier today. No chest pain or shortness of breath. Vital signs are stable with a temperature of 99.1, heart rate 83, respiratory rate 16, oxygen saturation 97%, and blood pressure 140/71. Patient remains awake and alert.

Text variant: brief_radio_missing_details

54-year-old male with history of cirrhosis and hepatic encephalopathy, presenting with persistent epigastric pain radiating around the abdomen. Vomited this AM, possible blood noted. Vitals stable, BP 140/70, O2 sat 97%, no respiratory distress. Transporting alert, pain ongoing.

POST-PROCESSING AND LEAKAGE REDUCTION

After generation, reports were scanned for hospital-based or leakage-prone wording.

DETECTED PATTERNS

- Imaging/lab results
- Specialist consult wording
- Hospital workflow phrases

METHOD

Regex-based candidate detection → minimal LLM rewrite → second-pass check.

RESULTS

- Round 1 candidates: 461 reports (5.39%)
- Round 2 remaining candidates: 168 reports (1.96%)

POST-PROCESSING AND LEAKAGE REDUCTION

The goal was to preserve clinical meaning while keeping the text EMS-style.

--- ORIGINAL TEXT ---

51-year-old male with history of esophageal cancer post-esophagectomy, presenting with 2 days of worsening crampy abdominal pain and bilious vomiting. CT shows small bowel obstruction. Vitals stable with BP 102 systolic, O2 sat 95% on room air. Transporting now.

--- CLEANED TEXT ---

51-year-old male with history of esophageal cancer post-esophagectomy, presenting with 2 days of worsening crampy abdominal pain and bilious vomiting consistent with small bowel obstruction. Vitals stable with BP 102 systolic, O2 sat 95% on room air. Transporting now.

TRAIN / VALIDATION / TEST SPLIT

Split strategy:

- Group-based split by source_case_id: all 4 variants of the same clinical case were kept in the same split
- Class coverage check: all target classes are represented in train, validation, and test.

This prevents leakage between train, validation, and test. **The same group-based split was preserved after ASR generation.*

Final split:

70% TRAIN:

5,988 reports (1,497 source cases)

15% VALIDATION:

1,284 reports (321 source cases)

15% TEST:

1,284 reports (321 source cases)

ASR ROBUSTNESS PIPELINE

Example of audio after
adding noise

GOAL

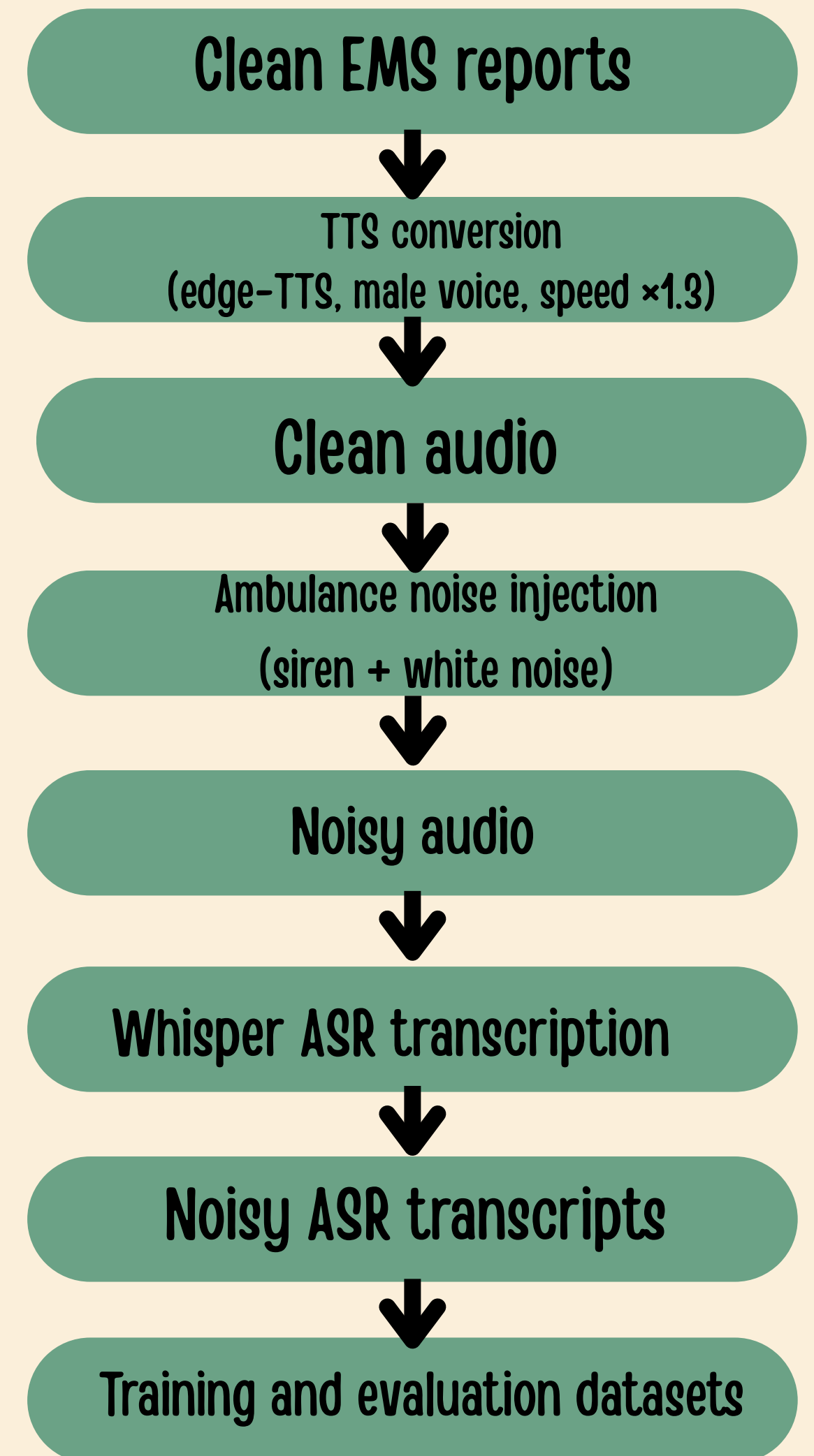
Test whether training with noisy ASR transcripts improves model robustness.

Training strategies:

- Clean text only
- Noisy ASR text only
- Clean + noisy ASR text

Evaluation:

The original source-case split was preserved, and the same noisy test set was used for all comparisons.



ASR Error Example: Clean Report vs Noisy Transcript

Mean WER: 0.443 $\approx$ 44%

WER measures the word-level difference between the clean EMS report and the ASR transcript.

noise dataset:
Average WER:
Train: 44.3%
Validation: 43.9%
Test: 44.8%

Original cleaned text:

We are transporting a 46-year-old female with a history of Crohn's disease who is experiencing sudden onset upper abdominal cramping radiating to her chest and back, consistent with her past Crohn's flares. She reports nausea and four episodes of bilious vomiting with small blood specks but denies diarrhea or constipation. Her vitals are stable: heart rate 72, blood pressure 132/65, respiratory rate 16, oxygen saturation 100%, and temperature 97.4. She has not taken her Humira injection in over two weeks. Patient is currently stable.

ASR noisy transcript:

We are transporting a 46-year-old female with a history of proven disease who was experiencing sudden onset of her abdominal cramping radiant or chest and back, consistent with her past crows' flares. She reports nausea and four episodes of billis fondly with small blood spests of deni-ciaria or constipation. Her vitals are stable, heart rate 72, blood pressure 152-165, risk occurring rate 16, oxygen saturation 100%, and temperature 97.4. She has not taken her cumulative injection in over two weeks. Patient is currently stable.

MODELS AND TRAINING STRATEGIES

EXPERIMENTAL STAGES

DistilBERT:

- General-purpose language model
- Smaller and faster baseline
- Included to test whether a general model maybe less sensitive to ASR errors

BioClinicalBERT:

- Pre-trained on biomedical and clinical text
- Better suited to medical terminology
- Used as the main domain-specific model

Stage 1 – Clean Baseline

- Train on clean text
- Test on clean text
- Establish the best baseline model
- for each target

Stage 2 – ASR Robustness Evaluation

- Train clean → Test noisy
- Train noisy → Test noisy

**Clean + noisy training → Noisy test
(BioClinicalBERT only)*

BEST BASELINE MODELS ON CLEAN TEXT

Optimization experiments

DistilBERT vs. BioClinicalBERT / Full fine-tuning vs. partial freezing / Weighted vs. unweighted loss / 3 vs. 5 epochs / Learning rate: 2e-5 vs. 1e-5 / Max length: 128 vs. 192

1. ED Care Area

Best model: BioClinicalBERT

- Partial fine-tuning: last 8 layers trainable
- Max length: 128
- 3 epochs
- Learning rate: 2e-5
- No class weights

Clean test results

Accuracy: 67.1%

Macro F1: 0.319

Weighted F1: 0.652

2. Primary Specialty Consultation

Best model: BioClinicalBERT

- Full fine-tuning
- Max length: 192
- 3 epochs
- Learning rate: 2e-5
- No class weights

Clean test results

Accuracy: 65.7%

Macro F1: 0.506

Weighted F1: 0.640

BEST DISTILBERT BASELINE MODELS ON CLEAN TEXT

Selected configuration

Full fine-tuning | No class weights | 3 epochs | Learning rate: 2e-5 | Max length: 128

ED Care Area

Clean test results:

Accuracy: 66.5%

Macro F1: 28.0%

Weighted F1: 63.8%

Primary Specialty Consultation

Clean test results:

Accuracy: 60.7%

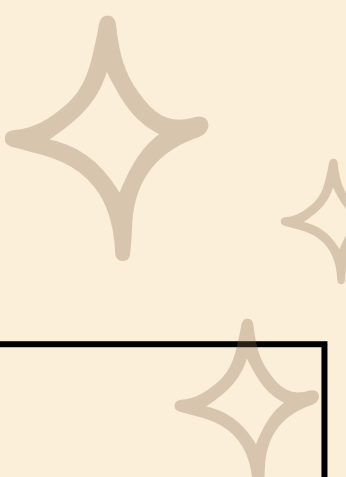
Macro F1: 34.4%

Weighted F1: 57.6%

1. CLEAN TRAINING → NOISY TEST

Model	Target	Clean Test Macro F1	Noisy Test Macro F1	Relative Change in Macro F1 (%)
DistilBERT	ED care area	28.00%	23.87%	-14.75%
BioClinicalBERT		31.86%	22.55%	-29.22%
DistilBERT	Specialty consultation	34.40%	22.45%	-34.74%
BioClinicalBERT		50.60%	27.74%	-45.18%

1. CLEAN TRAINING → NOISY TEST



Model	Target	Clean Test Accuracy	Noisy Test Accuracy	Change
DistilBERT	ED care area	66.50%	55.90%	-10.6 pp
BioClinicalBERT		67.10%	59.00%	-8.0 pp
DistilBERT	Specialty consultation	60.70%	40.80%	-19.9 pp
BioClinicalBERT		65.70%	39.30%	-26.4 pp

2. TRAIN NOISY ASR → TEST NOISY ASR

 Model	Target	Clean → Noisy Macro F1	Noisy → Noisy Macro F1	Relative Change in Macro F1
BioClinicalBERT	ED care area	22.55%	27.64%	22.57%+
DistilBERT		23.87%	21.98%	-7.92%
BioClinicalBERT	Specialty consultation	27.74%	31.21%	12.51%+
DistilBERT		22.45%	19.82%	-11.71%

Noisy training improved BioClinicalBERT in both targets, but did not improve DistilBERT according to Macro F1.

2. TRAIN NOISY ASR → TEST NOISY ASR



Model	Target	Clean → Noisy Accuracy	Noisy → Noisy Accuracy	Change
BioClinicalBERT	ED care area	59.03%	64.64%	+5.61 pp
DistilBERT		55.92%	62.46%	+6.54 pp
BioClinicalBERT	Specialty consultation	39.25%	49.77%	+10.52 pp
DistilBERT		40.81%	45.09%	+4.28 pp

EXAMPLE: NOISY TRAINING IMPROVED ROBUSTNESS

```
=====
EXAMPLE: NOISY TRAINING CORRECTED THE PREDICTION
=====
```

```
True specialty: orthopedic_consult
Clean-trained model prediction: ent_or_eye_oral_consult
Noisy-trained model prediction: orthopedic_consult
WER: 64.91%
```

```
Clean report
```

```
-----
We're bringing in a 27-year-old male who slipped on stairs carrying laundry and fractured
his left distal tibia. He denies passing out or hitting his head, no other pain, numbness,
or tingling reported. He's holding the ankle due to pain but no movement. Vitals are
stable: pulse 80, breathing 16, BP 140/70, oxygen 98% on room air.
```

```
noise ASR transcript
```

```
-----
We're bringing in a 27-year-old nail whose lip comes there to carry one re-intruch of his
leftus with tidius. He denies passing out or hitting his head, no other being, none as
once in the world reported. He's holding the ankle through the head, but no moving. Witals
are stable, plus 80, breathing 16, if he wants 40, slightly 70, oxygen 98% of the room
air.
```

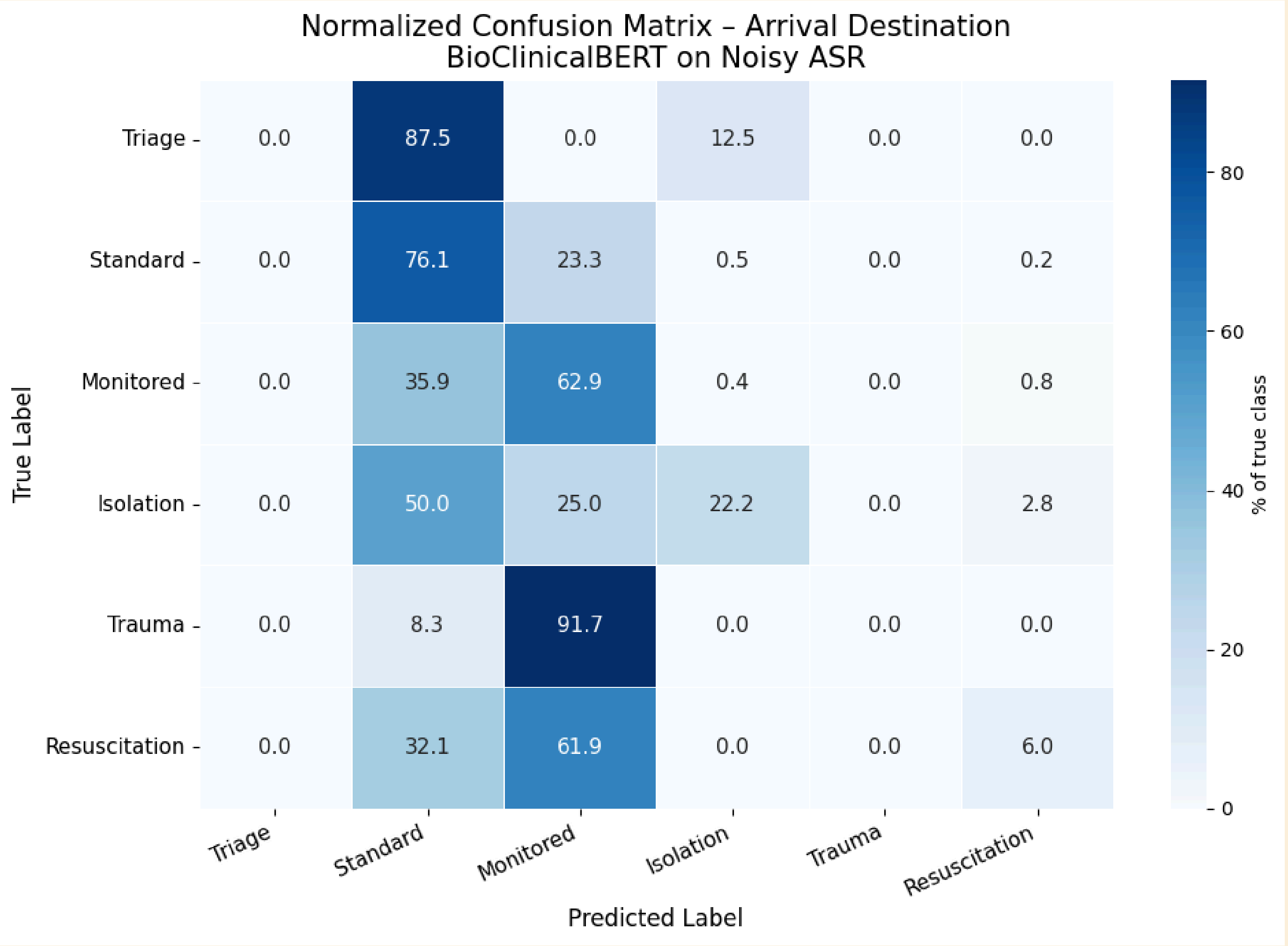
3. CLEAN + NOISY TRAINING → NOISY TEST:

BioClinicalBERT

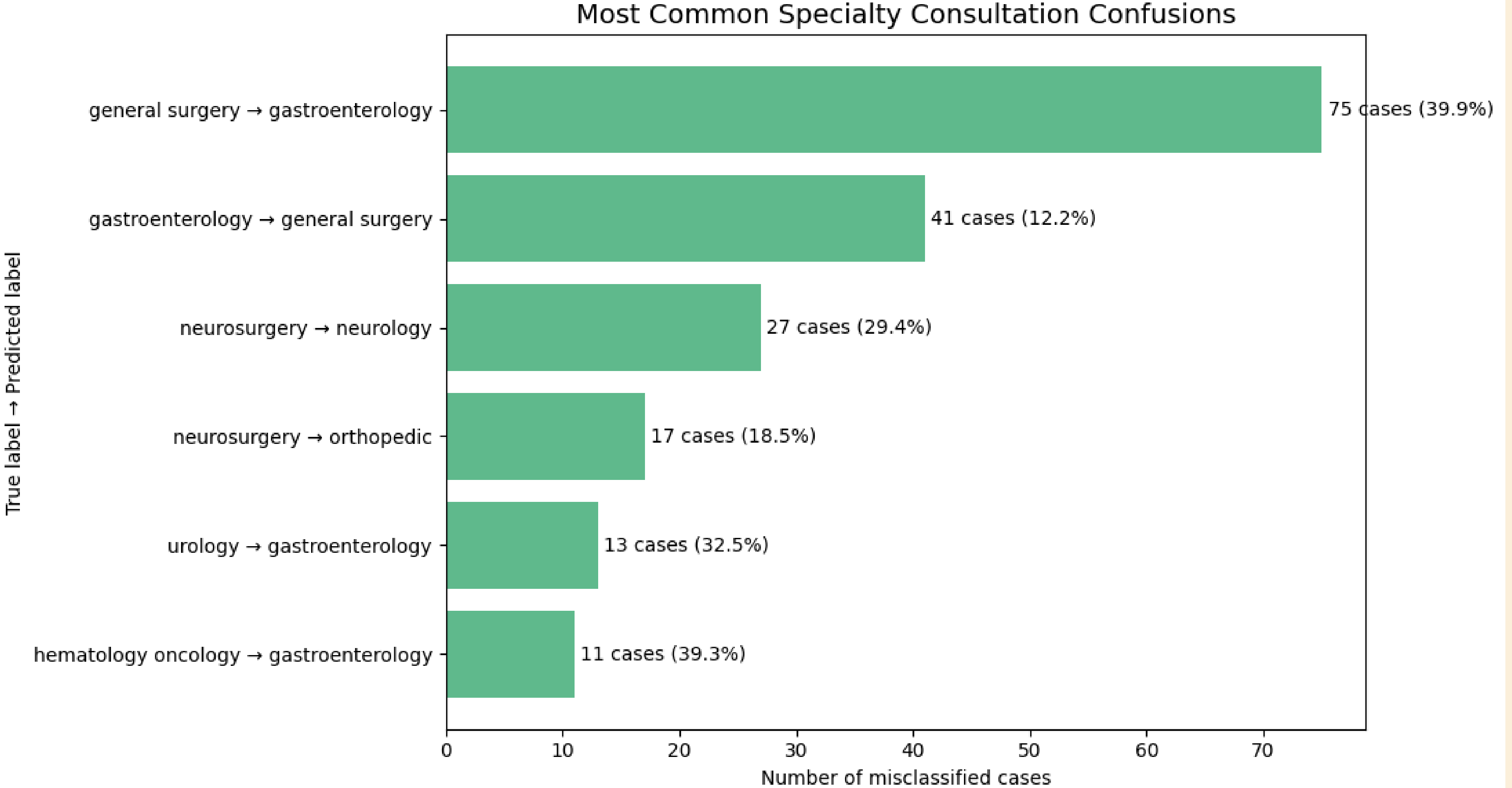
Target	Clean → Noisy Macro F1	Noisy → Noisy Macro F1	Clean + Noisy → Noisy Macro F1
ED care area	22.55%	27.64%	29.16%
Specialty consultation	27.74%	31.21%	38.33%

Target	Clean → Noisy Accuracy	Noisy → Noisy Accuracy	Clean + Noisy → Noisy: Accuracy
ED care area	59.03%	64.64%	63.63%
Specialty consultation	39.25%	49.77%	52.50%

Error Analysis



Error Analysis



Error Analysis

11.48% of the apparent errors were valid secondary specialties

```
=====
VALID SECONDARY PREDICTION IN THE DATASET ANNOTATION
=====

Case summary:
48-year-old male with diabetes, spreading gangrene and infection in the right
foot, with gas in the tissues and need for urgent surgical intervention.

Primary true label:
General Surgery

Model prediction:
Vascular Surgery

Valid specialty annotations:
General Surgery | Orthopedic | Vascular Surgery

Why is this important?
The prediction was counted as an error in the single-label evaluation, although
Vascular Surgery was marked as a valid consultation for this case.
=====
```


CONCLUSIONS AND FUTURE WORK



The solution improved robustness, but did not fully solve rare-class errors.

Performance was limited by ASR noise, class imbalance, label overlap, and partial multi-label ambiguity.

Oversampling and class-balanced training

More rare-class data

Multi-label specialty classification

Focus first on specialty consultation or ED care area, then extend to the second task.

Optimize the clean-to-noisy training ratio.